

# Physical AI SDK - Model Lifecycle & Strategy

2026

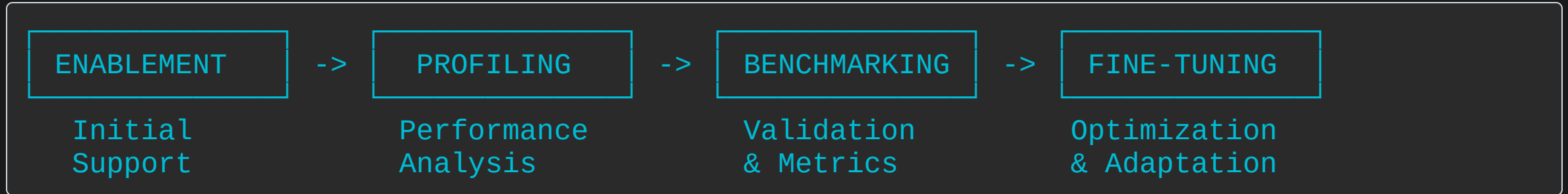
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# Model Lifecycle Overview

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## The Four-Stage Model Lifecycle



### Key Phases:

1. **Enablement** - Initial model support, integration, basic functionality
2. **Profiling** - Performance analysis, bottleneck identification, hotspot analysis
3. **Benchmarking** - KPI validation, performance metrics, competitive analysis
4. **Fine-tuning** - Model optimization, domain-specific adaptation, deployment ready

# Progressive Model Support Strategy

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## From Individual Models to System Orchestration

### Phase 1: Individual Model Support

- Focus on single model enablement and optimization
- Establish baseline performance for each model
- Build reusable infrastructure and tooling

### Phase 2: Multi-Model Orchestration

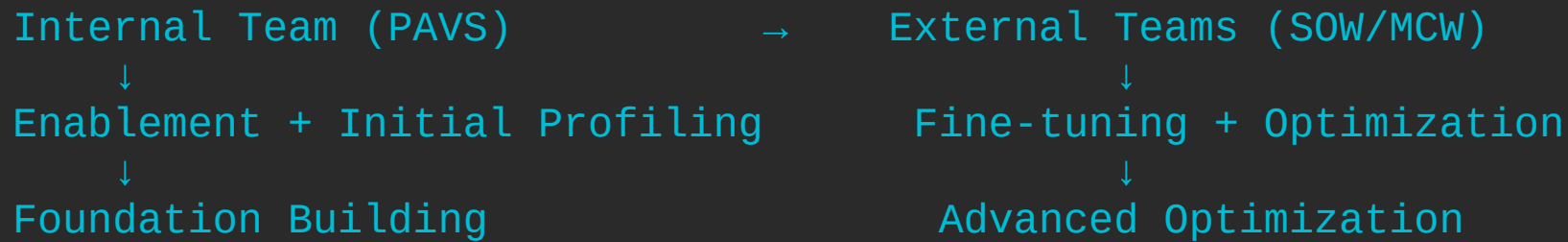
- Coordinate multiple models working together
- System-level optimization and resource management
- End-to-end pipeline integration

**Example Pipeline:** Vision → VLM → LLM → TTS

# Effective Team Collaboration

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## Progressive Engagement Model



### Strategy:

- **PAVS Team:** Handle enablement and initial profiling - establish working baseline, identify optimization opportunities
- **External Partners:** Focus on fine-tuning and advanced optimization - domain-specific optimization, large-scale benchmarking

**Benefits:** ✓ Better ROI ✓ Clear handoff criteria ✓ Faster time to production

# Speed Matters - Parallel Execution

## Current Model Status Across Lifecycle Stages

| Model Name      | Category               | Enablement | Profiling | Benchmarking | Fine-tuning |
|-----------------|------------------------|------------|-----------|--------------|-------------|
| MobileSAM       | Vision/Segmentation    | ✓          | ✓         | ✓            | 🔄           |
| YoloV12         | Object Detection       | ✓          | ✓         | ✓            | ⏸           |
| Qwen3-VL        | VLM                    | ✓          | ✓         | 🔄            | ⏸           |
| LLaMA3.2-Vision | VLM                    | ✓          | ✓         | ⏸            | ⏸           |
| Whisper         | ASR                    | ✓          | ✓         | ✓            | ✓           |
| XTTS            | TTS                    | ✓          | ✓         | ✓            | ⏸           |
| DeepSeek-R1     | LLM                    | ✓          | 🔄         | ⏸            | ⏸           |
| Qwen2.5-VL      | VLM                    | ✓          | ✓         | ✓            | 🔄           |
| CrossFormer     | Vision Transformer     | ✓          | 🔄         | ⏸            | ⏸           |
| CenterPoint     | 3D Detection           | ✓          | ⏸         | ⏸            | ⏸           |
| OpenVLA         | Vision-Language-Action | ✓          | ⏸         | ⏸            | ⏸           |
| EfficientNetV2  | Image Classification   | ✓          | ✓         | ✓            | ⏸           |

Legend: ✓ Complete | 🔄 In Progress | ⏸ Not Started

# Environment & CI/CD Infrastructure

## Isolated Environments for Each Model/System

### Environment Strategy:

```
physical-ai-sdk/  
├── models/  
│   ├── mobilesam/  
│   │   ├── Dockerfile  
│   │   ├── pyproject.toml (uv)  
│   │   └── .venv/  
│   └── yolov12/  
│       ├── Dockerfile  
│       ├── requirements.txt  
│       └── .venv/
```

**Benefits:** ✓ Isolated dependencies ✓ No version conflicts ✓  
Easy model-specific exploration

### Branch Strategy:

- `feature/*` → Developer branches
- `dev` → **Continuous work branch**
- `prod` → **Stable release branch**

### Model Deployment Flow:

1. Developer → Push to `dev`
2. CI/CD runs automated tests
3. Validated → Pull to `prod`
4. Final validation suite
5. Deploy to customers

# Container & Virtual Environment Options

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## Deployment Flexibility

### Option 1: Dockerfile

```
FROM rocm/pytorch:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install -r \
    requirements.txt
COPY . .
CMD ["python", \
    "inference.py"]
```

### Option 2: uv + pyproject.toml

```
[project]
name = "mobilesam"
version = "1.0.0"
dependencies = [
    "torch>=2.0.0",
    "onnxruntime-rocm",
]
```

### Option 3: pip + requirements

```
torch==2.0.0
onnxruntime-rocm==1.17.0
opencv-python==4.8.0
```

# Expanding Model Portfolio

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## Staying Current with Latest Open Source Models

### Emerging Neural Alternatives:

-  **End-to-End Neural Pipelines** replacing traditional CV pipelines
  - Example: Neural SLAM replacing traditional SLAM
  - Example: Neural 3D reconstruction replacing classical methods
-  **Latest Model Categories to Track:**
  - Multimodal Foundation Models (GPT-4V alternatives)
  - Action-VLMs (RT-2, OpenVLA successors)
  - Neural Rendering (NeRF, Gaussian Splatting for robotics)
  - On-device Edge LLMs (< 3B parameters)

### Strategy:

1. Monthly scan of arxiv.org, HuggingFace, GitHub trending
2. Quick feasibility assessment (30-day cycle)
3. Add to roadmap if aligned with Physical AI use cases



# Benchmarking & Documentation Strategy

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## GitHub Pages Integration

### Documentation as Code:

```
docs/
├── index.md
├── models/
│   ├── mobilesam.md
│   ├── yolov12.md
│   └── benchmarks/
│       ├── robotics.md
│       └── industrial.md
├── guides/
│   ├── quick-start.md
│   └── deployment.md
└── _config.yml
```

### Automated Workflow:

1. **Code + Benchmarks** → GitHub Repo
2. **CI/CD** → Auto-generate docs, run benchmarks
3. **GitHub Pages** → Publish results automatically

### Benefits:

- ✓ Version-controlled documentation
- ✓ Automated benchmark updates
- ✓ Professional presentation
- ✓ Community collaboration

# Comprehensive Benchmarking Strategy

## Dual-Mode Benchmarking & Regression Testing

### 1. NPU-Disabled Mode (Competitive Comparison)

- **Purpose:** Fair comparison with NVIDIA/Intel
- **Hardware:** NPU disabled, iGPU + CPU execution
- **Use Case:** Match industry-standard metrics

### 2. NPU-Enabled Mode (AMD Advantage)

- **Purpose:** Demonstrate AMD's hybrid execution
- **Hardware:** NPU + iGPU + CPU (full APU)
- **Use Case:** Showcase AMD performance gains

### Performance Comparison:

| Model     | NPU-Off (vs Competitors) | NPU-On (AMD Advantage) | Gain |
|-----------|--------------------------|------------------------|------|
| MobileSAM | 180 fps (iGPU)           | 240 fps (NPU)          | +33% |
| YoloV12   | 200 fps (iGPU)           | 240 fps (NPU)          | +20% |
| Whisper   | 85 tok/s (CPU)           | 100 tok/s (NPU)        | +18% |
| Qwen3-VL  | 25 fps (iGPU)            | 35 fps (NPU+iGPU)      | +40% |

# CI/CD Pipeline Detail

## Preventing Regressions



# Practical Example - Model Lifecycle

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## Case Study: Qwen3-VL Integration

### Week 1-2: Enablement

-  Model weights download and conversion
-  Basic ONNX-RT integration
-  Simple inference test (single image)
-  Docker environment setup

### Week 3-4: Profiling

-  Layer-wise performance analysis (AMD Profiler)
-  Memory usage tracking
-  Hotspot identification (Conv3D, GEMM)

**Result:** 2.5x speedup, 92% accuracy maintained

### Week 5-6: Benchmarking

-  KPI testing: Target 30fps @ 1080p
-  Accuracy validation on VQA datasets
-  Power consumption measurement

### Week 7-8: Fine-tuning

-  Triton kernel optimization for GEMM
-  INT8 quantization (W8A8)
-  Model-specific automotive dataset fine-tuning

# Resource Management Strategy

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## Balancing Internal & External Resources

### Resource Allocation Matrix:

| Lifecycle Stage | Internal Team (PAVS) | External Partners | Duration  |
|-----------------|----------------------|-------------------|-----------|
| Enablement      | Primary              | Support           | 2-3 weeks |
| Profiling       | Lead                 | Assist            | 2-4 weeks |
| Benchmarking    | Guide                | Execute           | 3-4 weeks |
| Fine-tuning     | Review               | Primary           | 4-8 weeks |

### Handoff Criteria:

- ✓ Model runs successfully on target hardware
- ✓ Profiling report completed
- ✓ Baseline benchmarks established
- ✓ Optimization opportunities documented

# Technology Watchlist 2026

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## Emerging Models & Technologies

### High Priority:

1. **Qwen3-VL 32B** - Latest multimodal foundation model
2. **RT-X** - Robotics transformer with broader action space
3. **Neural SLAM** - End-to-end learned SLAM systems
4. **Depth Anything V2** - Improved monocular depth

### Medium Priority:

5. **SAM 2** - Segment Anything with video support
6. **Florence-2** - Microsoft's vision foundation model
7. **Stable Diffusion 3** - For synthetic data generation
8. **Phi-3 Vision** - Compact VLM (< 4B params)

### Evaluation Criteria:

- ✓ Open source availability
- ✓ Relevance to Physical AI use cases
- ✓ Hardware compatibility (ROCm/NPU)
- ✓ Community adoption & maintenance

# Success Metrics & KPIs

## Measuring Lifecycle Effectiveness

### Model Lifecycle KPIs:

| Metric                        | Target         | Current   | Trend        |
|-------------------------------|----------------|-----------|--------------|
| Average Time to Enablement    | < 2 weeks      | 2.5 weeks | ⬇️ Improving |
| Models in Profiling Stage     | 5-8 concurrent | 6         | ➡️ Stable    |
| Benchmark Pass Rate (1st try) | > 80%          | 75%       | ⬆️ Improving |
| Fine-tuning Speedup (avg)     | > 2x           | 2.3x      | ⬆️ Good      |
| CI/CD Pipeline Success Rate   | > 95%          | 97%       | ✅ Excellent  |

**Portfolio Metrics:** Total: 45+ | Production-Ready: 28 | In Development: 12 | Experimental: 5

# Roadmap & Next Steps

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## 2026 Execution Plan

### Q1 2026 (Current):

-  Establish lifecycle framework
-  Complete 8 models through full lifecycle
-  Deploy GitHub Pages documentation

### Q2 2026:

-  Expand to 15 production-ready models
-  Implement automated benchmarking CI/CD
-  Onboard 2 external partner teams (MCW/SOW)

### Q3 2026:

-  Multi-model orchestration (5 system pipelines)
-  Add 10 emerging models from watchlist
-  Scale fine-tuning operations

### Q4 2026:

-  Full production deployment for 3 customer segments
-  Documentation portal v2.0
-  2027 strategy planning



# Call to Action

Building the Future of Physical AI

## Our Commitment:

🎯 **Systematic** approach to model support | 🎯 **Efficient** collaboration with partners  
🎯 **Rapid** adaptation to emerging technologies | 🎯 **Quality** focus through CI/CD

## Get Involved:

✉️ [zhaohui.yan@amd.com](mailto:zhaohui.yan@amd.com) | 💬 Slack: #pavs-ai-sdk

Together, we enable Physical AI at scale.